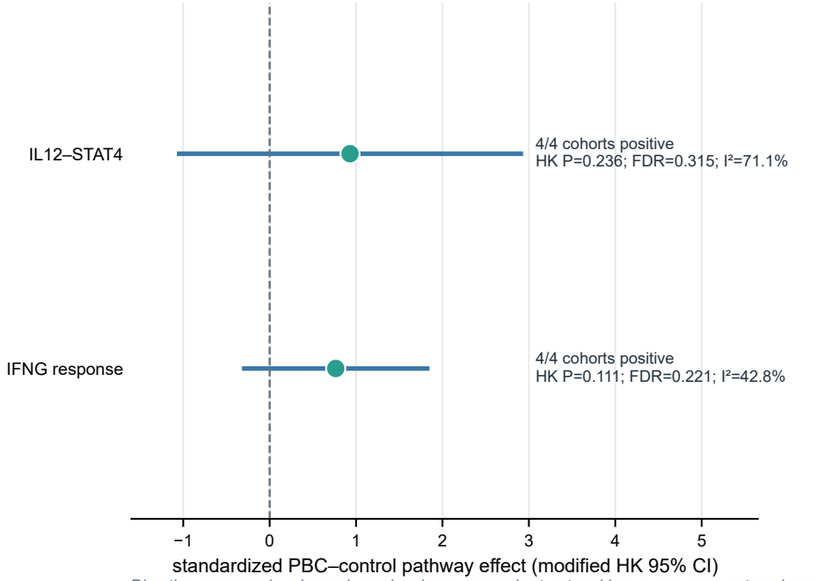
Orthogonal molecular evidence and a guarded translational path for the IL12RB2–TYK2 axis

Directional multi-omics support is separated from clinical efficacy, safety and druggability claims

A Bulk transcriptomic pathway meta-analysis

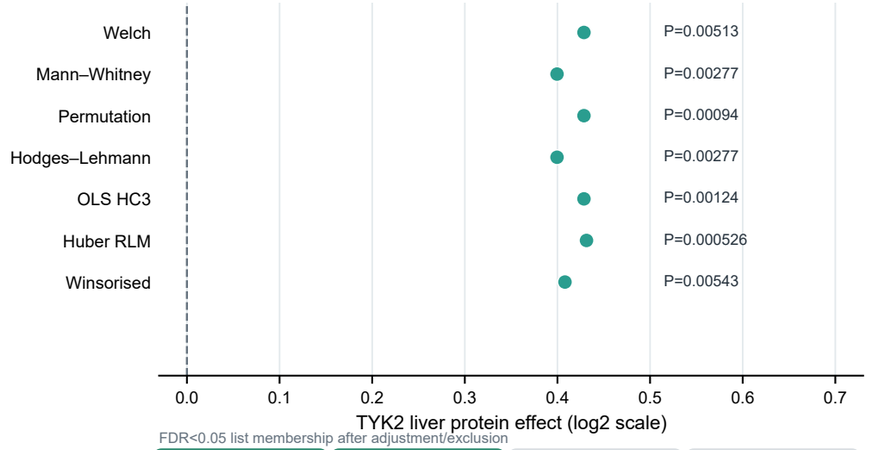
Four independent cohorts; prespecified robust-z score



Direction preserved under rank scoring, leave-one-cohort-out and leave-one-gene-out analyses; neither pathway reached multiplicity-adjusted significance.

B PXD083842 proteomic robustness and confounding

PBC n=14 versus healthy n=9



FDR<0.05 list membership after adjustment/exclusion

Unadjusted	**Cirrhosis**	Age	Sex
BMI	Age+sex	No MASLD	vs MASLD

Critical: age has minimal overlap; recruitment stream is exactly confounded with disease. Adjusted non-significant coefficients were not released. Supportive, not disease-specific replication.

C Drug and clinical evidence matrix

Open Targets / ChEMBL / ClinicalTrials.gov / labels

	Target / mechanism	Clinical-stage modality	PBC trial evidence	Label / safety evidence	Genetic-direction match
IL12RB2 direct	+	—	—	—	+
TYK2 / deucravacitinib	+	+	—	!	+
IL12B / ustekinumab	+	+	+	!	!
JAK1/2 / baricitinib	+	+	—	—	!

+ available/supportive ! qualified risk — absent or negative

PBC counterevidence: ustekinumab 0/20 achieved week-12 ALP response; baricitinib trial terminated after n=2. No TYK2-directed PBC trial found.

D Safety and druggability gates

Green does not denote efficacy; it denotes passage of that specific gate

TYK2 / deucravacitinib

Mechanistic and genetic direction	Matched
Selective approved TYK2 modality	Available
Direct efficacy evidence in PBC	Absent
ALT / AST and DILI monitoring	Required
Serious infection and latent TB	Core gate
Child–Pugh C	Not recommended

CONDITIONAL GO

Preclinical / early proof-of-mechanism only

Direct IL12RB2 small-molecule route

Experimental receptor structures	8XRP / 8YI7
Ligand-bound small-molecule pocket	Absent
ChEMBL IL12RB2 activity record	Absent
Validated docking benchmark	Absent
Receptor-selective biologic concept	Research only

NO-GO for docking

Reassess if a validated pocket or ligand emerges

E Final evidence grade

Claim strength is capped by the weakest required evidence layer

1 Genetic target support — MODERATE–STRONG

IL12RB2 locus/eQTL convergence; TYK2 human loss-of-function direction

2 Translational hypothesis — CONDITIONAL

Bulk direction is consistent but non-significant; proteomic signal is confounded; selective TYK2 is testable

3 Clinical recommendation — NOT SUPPORTED

No direct TYK2 PBC trial; prior pathway trials negative/inconclusive; PBC-specific efficacy and safety remain unknown

Final claim: genetically supported target axis with a guarded experimental drug hypothesis

All estimates and decisions are read from locked project tables. Database presence denotes evidence availability, not therapeutic efficacy. Audits current to 14–16 October 2026.